

SPARE PART CATALOG

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Power. Passion. Partnership.

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

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Preface

This catalog is the basic reference document for ordering spare parts. Illustrations are provided for extensive assembly operations to assist trained personal in conducting such operations but do not claim to be complete. For better overview, even product components that are not available as spare parts are shown.

The individual components were taken from the assembly list.

Catalog structure

The catalog is divided into Function Groups (FG) and Function Group Segments (FGS) in ascending order.

The side contain the following information:

- **MODEL:** (Product Type)
- **SERIAL NO.:**
- **FG/FGS DESIGNATION**
- **FG/FGS NO.:**
- **BOM No.:**

Explanation of columns

- **ITEM:** Provides a cross-reference between illustration and text section.
- **PART NO.:** This is the valid number for ordering the part in question.
- **DESIGNATION:** The precise part name.
- **MODEL/TYPE** is an additional specification.
- **QTY.: (Quantity in Unit):** Indicates the quantity of this part required for the case in question. Quantities not stated are available on request.
- **UNIT:** The 3-character codes indicate: PC = Piece, KG= Kilogram, M= Meter, ROL= Roll, PAC= Pack
- **WEIGHT**
- **FN.:** (Footnote) Provides additional parts information. For clarification of footnote symbols, see Page 2.
- **DIMENSIONS:** This column contains standard-part dimensions or manufacturer designations for purchased parts.
- **STANDARD**

Amendments service

Part No. changes and amendments which occur after publication of the catalog are not published. We therefore recommend that users make a record of amendments in their catalogs. This information is provided, for instance, in order confirmations and/or on spare parts labels.

Footnotes: Key

AB	Only available complete. Repair by manufacturer only.
AC	Not scheduled as a spare part. Availability must be checked.
ACN	Not scheduled as a spare part. Availability must be checked. Component only available complete.
EM	Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.
EMN	Emission-relevant parts. Component only available complete.
KD	Group balancing required
KE	Individual part; not scheduled as a spare part.
KF	Adjusted spare part.
KFN	Adjusted spare part. Component only available complete.
M	Spares-related components or parts with different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.
MN	Spares-related components or parts with different repair sizes. Components only available complete.
MUB	Spares-related components or Parts with different repair sizes. Available only as complete set.
NN	Component is not broken down into further sub-components.
P	List of bought-in components
SB	Group of parts which are welded or brazed together
UA	Indicate dimensions when ordering
UB	Only supplied complete or as a complete set.

Ordering Procedure

The following information is essential for spare-part order processing:

- Engine Model
- Serial number (see engine name plate)
- FG/FGS
- Item
- Part No.
- Designation
- Model/Type
- Quantity
- Page

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

When ordering a calibrated part (marked by "KF", "KFN" in the "FN" column), type the following additional information in the **Comments** field:

If the calibration No. is known (see part's nameplate):

Add the calibration No., preceded by an oblique stroke, as an extension to the part number ("/xxxxx")

e.g.: 0025362810/E1234 Here, the extension "E1234" would be the part's calibration number on the nameplate.

If the calibration No. is not known:

Give a detailed description of the required part's installation location by quoting the complete Device Description code (e.g. "= 37.1.1 + A022 - F003") plus the part number extension "/77" (e.g. 5005362810/77).

In this case, the calibrated spare part will be supplied under a new part number.

The Device Description is found on the nameplate at the part's installation location, or in the corresponding Wiring Diagram.

When ordering a non-calibrated adjustable part (marked by "KF", "KFN" in the "FN" column) which should not be calibrated by the factory: Add extension "/76" to the part number (e.g. 0025362810/76).

Phone orders should be confirmed in writing to prevent errors.

Ordering Example:

Engine model	Serial number	FG/FGS	ITEM	Part No.	Designation	Quantity	Page
16V4000R41	527101836	031 88	5	5270300701	Crankshaft	1	33
		083 05	10	0020922801	Easy-Cange Filter	2	92
Blue Line	V4000M Waterjet	517 00	75	5185301593	Control Console	1	53

Factory repairs

Parts shipped to MTU for repair must be marked with the relevant engine serial number (tag or label, for example).

Exchange components (Reman)

Exchange components are parts overhauled by the manufacturer. They have the same service life and quality characteristics as new parts.

Components which are included in the Reman exchange program are marked by the following letters in front of the item number:

For instance: **E 524 010 3520 = New component in exchange program**

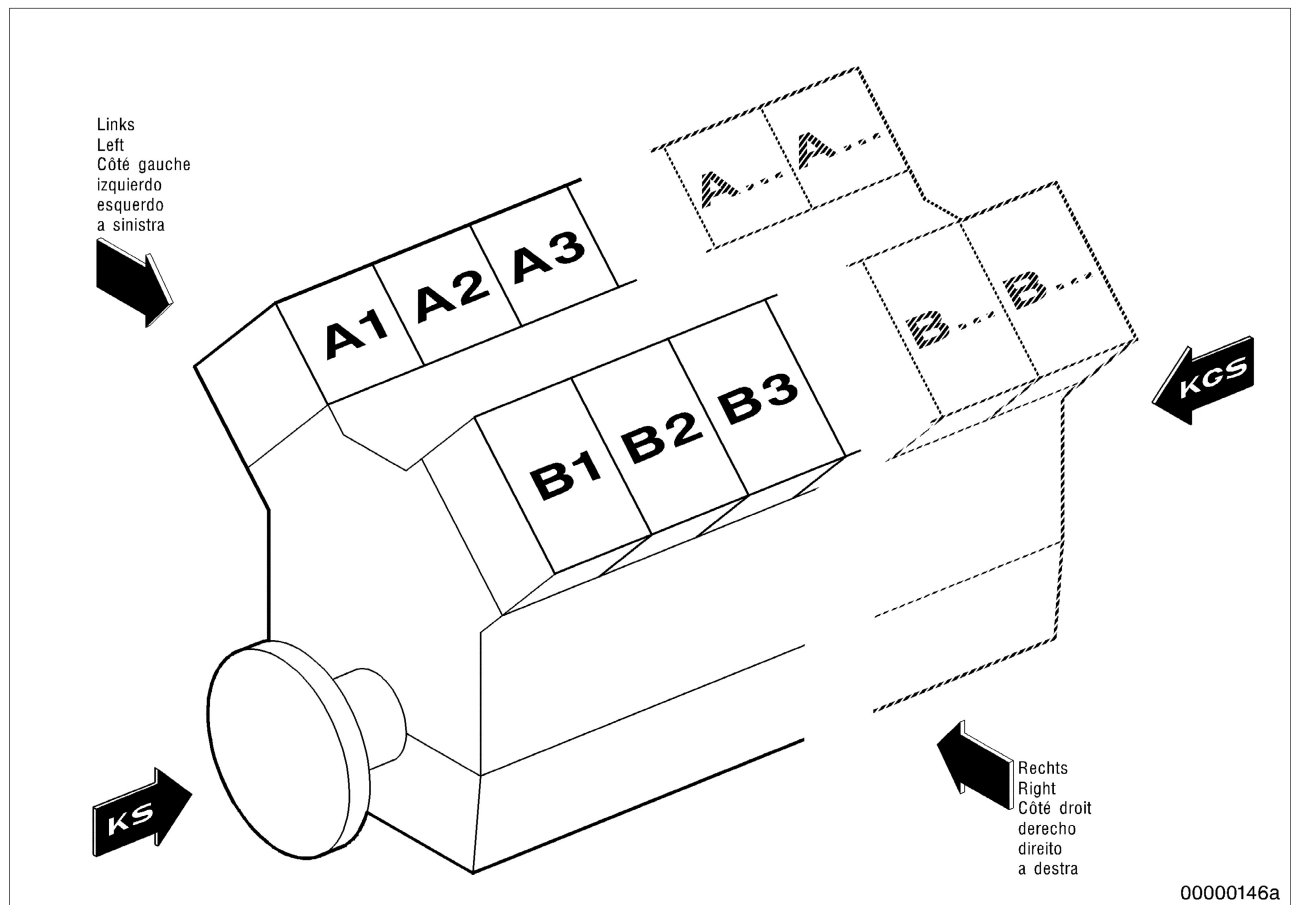
R 524 010 3520 = Reman component in exchange program

Exchange procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

Further information upon request.

Engine side and cylinder designations



Engine sides / banks are always designated as viewed from the driving end (KS).

The left bank of cylinders is designated "A" and the right bank "B" (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively from the driving end, starting with No. 1.

Other engine components are also numbered consecutively from the driving end, starting with No. 1.

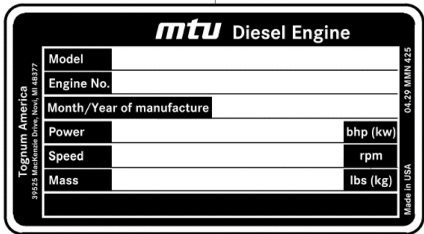
Designations and abbreviations

Driving end	= KS
Free end	= KGS
Left-hand side	
Right-hand side	
Top	
Bottom	
Turbocharger	= ATL

FG FG DESIGNATION
004 LABELS AND AUXILIARY MATERIALS
011 CRANKCASE
014 OIL PAN
015 ENGINE MOUNTING
016 END HOUSING, DRIVING END AND FREE END
018 CRANKCASE BREATHER
021 GEARCASE, DRIVING END
024 GEARCASE, AUX PTO END
031 CRANKSHAFT WITH BEARINGS
032 CRANKSHAFT, PTO ON DRIVING END
035 CRANKSHAFT, PTO ON AUX PTO END
037 PISTON
041 CYLINDER HEAD
051 CAMSHAFT
054 ROLLER TAPPETS, PUSH RODS
055 VALVE DRIVE
075 FUEL INJECTOR
080 L.P. FUEL SYSTEM; (ALSO GAS ENGINES)
081 FUEL PIPEWORK BEFORE DELIVERY PUMP
084 FUEL PIPEWORK FROM FILTER TO INJECTION PUMP / HP PUMP
101 EXHAUST TURBOCHARGER, GENERAL OR H.P. TURBOCHARGER
124 AIR SUPPLY TO CYLINDERS
141 EXHAUST SYSTEM AFTER CYLINDER HEAD
172 ELECTRIC STARTING
181 LUBE OIL PUMP AND DRIVE
183 OIL FILTRATION/COOLING
185 TURBOCHARGER OIL SUPPLY
188 CONNECTING HARDWARE FOR ENGINE OIL SUPPLY
202 COOLANT CIRCUIT, GENERAL, HT CIRCUIT
206 COOLANT PIPEWORK WITH THERMOSTAT
213 GENERATOR
221 MECHANICAL FAN DRIVE
231 ENGINE MOUNTING
507 WIRING, GENERAL, FOR ENGINE/GEARBOX/PLANT
820 PACKING OF COMPLETE ENGINE
XX XX

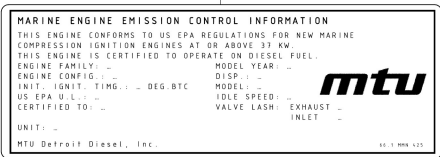
GROUP SEGMENT (FGS)
GROUP NAME
BOM-No.

05
TYPE PLATE USA ENGLISH
XS524000.00115



00400248

05
EMI-LABEL FOR US (RE-)EXPORT
XS524000.00022



00400098a